Chapter II: Hamiltonian Methods: Matter, Light & Interactions

3 Approximating Eigenstates of Quantum Hamiltonians

We will use the problem of bound states to understand the connection between Hamiltonian symmetries, degeneracies in the spectrum and the use/need for quantum numbers. This section will also give us practice in real space and momentum space techniques.

3.1 Bound states in real space: Variational Principle

3.1.1 Theoretical foundations

As we have seen through this course, unfortunately the number of quantum systems whose Hamiltonian admits an exact, analytic diagonalisation is quite scarce. This is why approximate methods are so hugely important in quantum mechanics. Here we are going to present a first of these approximate methods, known as the *variational principle*. The main idea is to find some test wave function that provides an upper bound to the energy of the ground state of the system. By means of a clever choice of this test wave function, together with an optimization of its adjustable parameters, we can obtain a very decent estimate of the ground state energy and wave functions of relatively complex systems without the need to solve the problem exactly.

The key result underlying the variational principle method is the following. We would to determine the ground state of a given Hamiltonian $\hat{H}$, but are unable to solve its eigenvalue equation exactly for the ground state (or for any other state in general):

$$\hat{H}\,|\psi_{\rm gs}\rangle = E_{\rm gs}\,|\psi_{\rm gs}\rangle \tag{213}$$

Else we need to resort to some approximate method to estimate the ground state energy $E_{\rm ga}$ of this quantum system. We can demonstrate that if we select at random some normalised state vector $|\psi\rangle$, then we can obtain an upper bound on the ground state energy by evaluating the following matrix element:

$$E_{\rm gs} \leq \langle\psi|\hat{H}|\psi\rangle = \langle\hat{H}\rangle \tag{214}$$

where the equality is only true if we use $|\psi_{gs}\rangle$ as our test function (at least, assuming that the ground state of our Hamiltonian is non-degenerate). In other words, the expectation value of the Hamiltonian of the system

on some test function is guaranteed to provide an upper bound on E_{gs}. This result is of course trivial if $|\psi\rangle$ is one of the excited states of the Hamiltonian, but perhaps less so if this state vector has been genuinely chosen at random.

Let us demonstrate that this result is indeed true. To show this, I will express my test state $|\psi\rangle$ as a linear combination of the eigenvectors of my Hamiltonian, that is

$$|\psi\rangle = \sum_n c_n |\psi_n\rangle \text{ where } \hat{H}|\psi_n\rangle = E_n |\psi_n\rangle \tag{215}$$

where as usual $\sum_n |c_n|^2 = 1$ due to the normalisation of the state vector, and the coefficients are computed as $c_n = \langle \psi_n \mid \psi \rangle$. I can now use this expansion to evaluate the expectation value of the Hamiltonian in the test state vector $|\psi\rangle$ as follows:

$$\langle \psi \mid \hat{H}\psi \rangle = \sum_{n,m} c_n^* c_m \langle \psi_m | \hat{H} | \psi_n \rangle = \sum_{n,m} c_n^* c_m E_n \langle \psi_m \mid \psi_n \rangle = \sum_n |c_n|^2 E_n \tag{216}$$

Again, this is what we expect: $\langle \psi \mid \hat{H}\psi \rangle$ is nothing but the expectation value of the energy in our state $|\psi\rangle$. However, the ground state energy is by definition the lowest of the energy eigenvalues E_n, and hence

$$\langle \psi | \hat{H} | \psi \rangle = \sum_n |c_n|^2 E_n \geq \sum_n |c_n|^2 E_{\text{gs}} = E_{\text{gs}} \tag{217}$$

thus demonstrating the main result of the variational principle. As you might have guessed, clever choices of the test state vector will provide much better estimates to E_{gs} than genuinely random choices.

3.1.2 The variational method for the quantum harmonic oscillator

Let us now demonstrate how this method works for one specific quantum state for which we know the exact solution, and hence we can check whether or not the estimate provided by the variational principle is reasonable. Let's consider the quantum harmonic oscillator and try a Gaussian test function

$$\psi(x, b) = Ae^{-bx^2} \tag{218}$$

where A is determined by normalisation and b is some positive real constant, which plays the role of adjustable parameter in our test function. You can check that $A = (2b/\pi)^{1/4}$ by evaluating the usual definite Gaussian integral, so b is the only genuine free parameter of our test function for the ground state of the system.

Following now the prescription provided by the variational principle, we need to take the Hamiltonian associated to this system

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d}{dx^2} + \frac{1}{2}m\omega^2 x^2 \tag{219}$$

and then compute its expectation value in the trial state defined in Eq. (218). We can evaluate separately the contributions from the kinetic and from the potential energy and find:

$$\begin{aligned} \langle T \rangle &= -\frac{\hbar^2}{2m}\left(\frac{2b}{\pi}\right)^{1/2}\int_{-\infty}^{\infty} dx e^{-bx^2}\left(\frac{d^2}{dx^2}e^{-bx^2}\right) = \frac{\hbar^2 b}{2m} \\ \langle V \rangle &= \frac{1}{2}m\omega^2\left(\frac{2b}{\pi}\right)^{1/2}\int_{-\infty}^{\infty} dx e^{-bx^2}\left(x^2 e^{-bx^2}\right) = \frac{m\omega^2}{8b} \end{aligned} \tag{220}$$

and combining the two contributions we find that an upper limit to the energy of the ground state of the harmonic operator Hamiltonian is given by

$$\langle H \rangle(b) = \left(\frac{\hbar^2 b}{2m} + \frac{m\omega^2}{8b}\right) \geq E_{\rm gs} \tag{221}$$

Now, a crucial observation is that this result holds true for any value of the free parameter b, and hence the best estimate that we can have for the ground state energy is obtained by minimizing this expectation value of $\hat{H}$ with respect to the value of b :

$$\frac{d\langle H \rangle(b)}{db} = \frac{\hbar^2}{2m} - \frac{m\omega^2}{8b^2} = 0 \quad \rightarrow \quad b = \frac{m\omega}{2\hbar} \tag{222}$$

We can now collect our results: our best estimate for the wave function and energy of the ground state of the quantum harmonic oscillator and its energy obtained by virtue of the variational principle (that is, we did not have to solve any differential equation!) will be

$$\psi(x) = \left(\frac{2b}{\pi}\right)^{1/4}\exp\left(-\frac{m\omega}{2\hbar}x^2\right), \quad E_{\rm gs} \leq \frac{1}{2}\hbar\omega \tag{223}$$

By comparing with the exact solution for the ground state wave function of the harmonic oscillator, we see that in this case the equality is satisfied and the resulting wave functions and energies are identical to those of the exact result. This is no coincidence, since we have chosen a test function which is Gaussian, which is the same as in the exact solution. If you would have tried a different test function, such as for example

$$\psi(x, b) = Ae^{-b|x|} \tag{224}$$

we would not have saturated the upper bound on $E_{\rm gs}$, as you can check explicitly [Workshop Exercise].

3.1.3 Choosing the test function

As we have seen from this example, the closer our test function $|\psi\rangle$ is to the actual ground state of the system $|\psi_{\text{gs}}\rangle$, the better the estimate provided by the variational principle will be. Actually, if you choose a functional form which is consistent with $\psi_{\text{gs}}(x)$ and then adjust the tunable parameters, you are guaranteed to recover the exact result for the ground state wave function $\psi_{gs}(x)$ and the energy E_{gs} without having to solve the Schroedinger equation.

3.2 Bound states in momentum space: Numerical diagonalisation

Fourier Transform: Definitions and useful properties

In d-dimensional space, the Fourier transform of a function $f(\mathbf{r})$ is defined as

$$\tilde{f}(\mathbf{k}) = \int_{\mathbb{R}^d} f(\mathbf{r})\, e^{-i\mathbf{k}\cdot\mathbf{r}}\, d^d\mathbf{r}, \tag{225}$$

with inverse transform

$$f(\mathbf{r}) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \tilde{f}(\mathbf{k})\, e^{i\mathbf{k}\cdot\mathbf{r}}\, d^d\mathbf{k}. \tag{226}$$

The Dirac delta function is defined by the integral

$$\delta^{(d)}(\mathbf{r}) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{i\mathbf{k}\cdot\mathbf{r}}\, d^d\mathbf{k}, \tag{227}$$

and satisfies the sifting property

$$\int_{\mathbb{R}^d} \delta^{(d)}(\mathbf{r}-\mathbf{r}')\, f(\mathbf{r}')\, d^d\mathbf{r}' = f(\mathbf{r}). \tag{228}$$

Convolution Theorem

The convolution of two functions $f(\mathbf{r})$ and $g(\mathbf{r})$ is defined as

$$(f * g)(\mathbf{r}) = \int_{\mathbb{R}^d} f(\mathbf{r}')\, g(\mathbf{r}-\mathbf{r}')\, d^d\mathbf{r}'. \tag{229}$$

The Fourier transform of a convolution is the product of the Fourier transforms:

$$\mathcal{F}[f * g](\mathbf{k}) = \tilde{f}(\mathbf{k}) \cdot \tilde{g}(\mathbf{k}). \tag{230}$$

Conversely, the Fourier transform of a product becomes a convolution:

$$\mathcal{F}[f(\mathbf{r})\, g(\mathbf{r})](\mathbf{k}) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \tilde{f}(\mathbf{q})\, \tilde{g}(\mathbf{k}-\mathbf{q})\, d^d\mathbf{q}. \tag{231}$$

Theory outline

Considering the two-body Hamiltonian $\hat{H}(\boldsymbol{r})$ in real space, we Fourier transform to momentum space and solve the eigenvalue problem. This lends itself to a direct numerical treatment.

We take as a particular example the single-particle Hamiltonian in 2D:

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\mathbf{r}), \tag{232}$$

with a central potential

$$V(\mathbf{r}) = \frac{1}{r}. \tag{233}$$

Step 1: Fourier transform the Schrödinger equation

Wavefunction

We define the momentum-space wavefunction

$$\tilde{\psi}(\mathbf{k}) = \int d^2\mathbf{r}\, e^{-i\mathbf{k}\cdot\mathbf{r}} \psi(\mathbf{r}). \tag{234}$$

Potential

Fourier transforming the potential gives

$$\tilde{V}(\mathbf{q}) = \int d^2\mathbf{r}\, e^{-i\mathbf{q}\cdot\mathbf{r}} \frac{1}{|\mathbf{r}|} \tag{235}$$

$$= \int r dr d\theta\, e^{-iqr\cos\theta} \frac{1}{r} \tag{236}$$

$$= \int dr\, 2\pi J_0(qr) = \frac{2\pi}{|\mathbf{q}|}. \tag{237}$$

Convolution

Using the convolution theorem:

$$\mathcal{F}\left[V(\mathbf{r})\psi(\mathbf{r})\right](\mathbf{k}) = \int \frac{d^2\mathbf{k}'}{(2\pi)^2} \tilde{V}(\mathbf{k}-\mathbf{k}')\tilde{\psi}(\mathbf{k}'). \tag{238}$$

Kinetic term

We derive the kinetic term in Fourier space:

$$\mathcal{F}\left[\nabla^2\psi(\boldsymbol{r})\right](\mathbf{k}) = \int d^2\mathbf{r}\, e^{-i\mathbf{k}\cdot\mathbf{r}}\nabla^2\psi(\boldsymbol{r}) \tag{239}$$

$$= \int \frac{d^2q}{(2\pi)^2}(i\boldsymbol{q})^2(2\pi)^2\delta^{(2)}(\boldsymbol{q}-\boldsymbol{k})\tilde{\psi}(\boldsymbol{q}) \tag{240}$$

$$= -k^2\tilde{\psi}(\boldsymbol{k}). \tag{241}$$

Thus, the kinetic energy operator becomes multiplicative in momentum space:

$$\hat{T} \to \frac{\hbar^2k^2}{2m}. \tag{242}$$

Step 2: Schrödinger equation in momentum space – Convert to a numerical matrix

The full Schrödinger equation becomes:

$$\frac{\hbar^2k^2}{2m}\tilde{\psi}(\mathbf{k}) + \int \frac{d^2\mathbf{k}'}{(2\pi)^2}\,\tilde{V}(\mathbf{k}-\mathbf{k}')\tilde{\psi}(\mathbf{k}') = E\tilde{\psi}(\mathbf{k}). \tag{243}$$

We now discretise momentum space as $\boldsymbol{k} \to \boldsymbol{k}_i$ with step size Δk. The integral becomes a sum (here we take a 2D example):

$$\int d^2\mathbf{k} = \int dk_x dk_y \to \sum_j \Delta k^2. \tag{244}$$

This gives a discrete eigenvalue problem:

$$\sum_j \delta_{ij}\frac{\hbar^2k_j^2}{2m}\tilde{\psi}(\mathbf{k}_j) + \sum_j \frac{\Delta k^2}{(2\pi)^2}\,\tilde{V}(\mathbf{k}_i-\mathbf{k}_j)\tilde{\psi}(\mathbf{k}_j) = E\tilde{\psi}(\mathbf{k}_i). \tag{245}$$

which can be written more explicitly as a matrix equation

$$H_{ij}\tilde{\psi}_j = E\tilde{\psi}_i. \tag{246}$$

Step 3: Numerically solve

Numerically filling the matrix elements H_{ij} is, in generally, computationally inexpensive. Subsequently finding the eigenstates and eigenvalues, numerically, is even easier.

Numerical convergence

Taking smaller step size Δk and hence more k-points results in more accurate results. To check for convergnece one can plot the eieganvlues (mauybe the first few) as a function of step size, eventually for small enough Δk the eigenvalues won't change — convergence.

The total number of eigenstates equals the total number of k-points, since this defines this size of the Hamitlonian matrix.

3.3 Numerical Hamiltonian Methods: Applications to the Transmon

In the previous sections we developed two powerful techniques for studying quantum Hamiltonians. The variational principle provides approximate solutions for ground states, while numerical diagonalisation allows us to obtain essentially exact eigenvalues and eigenstates of arbitrary Hamiltonians. Although these techniques were introduced in the context of particles moving through real space, they are considerably more general. The only ingredients required are a Hamiltonian and a pair of conjugate variables.

In this section we apply these Hamiltonian methods to one of the most important quantum devices used today: the superconducting transmon qubit. Superconducting qubits form the basis of many quantum computing platforms, including those developed by IBM and Google. Rather than describing the motion of a particle through space, the transmon is described by two different dynamical variables, namely the superconducting phase and the number of Cooper pairs that have crossed a Josephson junction.

The emphasis here is not on the microscopic origin of the transmon Hamiltonian, which is covered in *Solid-State Physics and Quantum Devices*, but rather on treating it as a quantum Hamiltonian that can be analysed using exactly the same methods developed earlier in this chapter. In doing so we will see that many familiar ideas from quantum mechanics — harmonic oscillators, Fourier transforms, numerical diagonalisation and matrix representations — reappear in a new physical setting.

3.3.1 Canonical Variables

Throughout this chapter we have worked with the familiar canonical variables position, x, and momentum, $\hat{p}$, satisfying the commutation relation

$$[x, \hat{p}] = i\hbar. \tag{247}$$

These variables are not unique. Quantum mechanics may equally well be formulated using any pair of conjugate variables. What makes two variables conjugate is not their physical interpretation, but rather that they satisfy a non-zero commutation relation.

For superconducting circuits the natural variables are

- the superconducting phase difference across a Josephson junction, denoted ϕ, and
- the number of Cooper pairs transferred across the junction, denoted $\hat{N}$.

These obey the commutation relation

$$[\phi, \hat{N}] = i. \tag{248}$$

Apart from the absence of the factor of $\hbar$, this relation is mathematically identical to that of position and momentum. Consequently, many of the methods developed earlier in this course carry over immediately.

The correspondence between the two systems is

Particle mechanics	Superconducting circuit
Position x	Phase ϕ
Momentum $\hat{p}$	Cooper-pair number $\hat{N}$
Potential $V(x)$	Josephson potential $-E_J \cos\phi$

This analogy will repeatedly guide our physical intuition throughout this section.

3.3.2 The Transmon Hamiltonian

A superconducting qubit consists of a small superconducting island connected to a larger superconducting reservoir through a Josephson junction. The dynamics are governed by two competing energy scales.

The first is the charging energy. Transferring Cooper pairs onto the superconducting island requires electrostatic energy, giving rise to the Hamiltonian

$$H_C = 4E_C\hat{N}^2, \tag{249}$$

where E_C is known as the charging energy.

The second contribution arises from coherent tunnelling of Cooper pairs across the Josephson junction. This lowers the energy according to

$$H_J = -E_J \cos\phi, \tag{250}$$

where E_J is the Josephson energy.

Combining the two contributions gives the transmon Hamiltonian

$$H = 4E_C\hat{N}^2 - E_J \cos\phi. \tag{251}$$

This remarkably compact Hamiltonian captures the essential physics of the transmon qubit. Although its microscopic origin involves superconductivity and Cooper-pair tunnelling, once written in the form of Eq. (251) it is simply another quantum Hamiltonian whose eigenvalues and eigenstates may be obtained using standard quantum mechanics.

3.3.3 Phase and Number Representations

Just as ordinary quantum mechanics admits both the position and momentum representations, the transmon Hamiltonian may be expressed in either the phase basis or the number basis.

Phase basis

Suppose the quantum state is represented by a wavefunction

$$\psi(\phi). \tag{252}$$

Because ϕ and $\hat{N}$ are conjugate variables, the number operator acts as a derivative,

$$\hat{N} = -i\frac{\partial}{\partial\phi}, \tag{253}$$

directly analogous to the momentum operator

$$\hat{p} = -i\hbar\frac{\partial}{\partial x}. \tag{254}$$

Substituting this expression into Eq. (251) gives

$$H = -4E_C\frac{\partial^2}{\partial\phi^2} - E_J\cos\phi. \tag{255}$$

Notice how similar this Hamiltonian is to those studied earlier in this chapter. The kinetic energy has become a second derivative with respect to ϕ, while the potential energy is simply the cosine potential.

Equation (255) could therefore be solved using exactly the same methods introduced previously for particles moving in one-dimensional potentials.

Physics insight. Although Eq. (255) resembles the Schrödinger equation of a particle moving through a periodic cosine potential, the variable ϕ is not a spatial coordinate. Instead it represents the superconducting phase difference across the Josephson junction. The mathematics is identical, but the physical interpretation is entirely different.

Number basis

An equally useful representation is obtained by introducing eigenstates of the number operator,

$$\hat{N}|N\rangle = N|N\rangle, \tag{256}$$

where the integer N counts the net number of Cooper pairs transferred across the junction.

In this basis the charging term is already diagonal,

$$4E_C\hat{N}^2 = 4E_C\sum_N N^2|N\rangle\langle N|. \tag{257}$$

The Josephson term, however, changes the number of transferred Cooper pairs by one. Using the Fourier relationship between phase and number, one finds

$$e^{\pm i\phi}|N\rangle = |N \pm 1\rangle. \tag{258}$$

Exercise: Show that this is true by showing that $\hat{N}(e^{\pm i\phi}\,|N\rangle) = (N \pm 1)(e^{\pm i\phi}\,|N\rangle)$ and hence that $(e^{\pm i\phi}\,|N\rangle) \equiv |N \pm 1\rangle$. You will need to use that $\hat{N} = -id/d\phi$ and $\hat{N}\,|N\rangle = N\,|N\rangle$.

Since

$$\cos\phi = \frac{1}{2}\left(e^{i\phi} + e^{-i\phi}\right), \tag{259}$$

the Josephson Hamiltonian becomes

$$H_J = -\frac{E_J}{2}\sum_N \left(|N\rangle\langle N+1| + |N+1\rangle\langle N|\right). \tag{260}$$

The full Hamiltonian in the number basis is therefore

$$H = 4E_C\sum_N N^2|N\rangle\langle N| - \frac{E_J}{2}\sum_N \left(|N\rangle\langle N+1| + |N+1\rangle\langle N|\right). \tag{261}$$

Several important observations follow immediately.

1. The charging energy is diagonal in the number basis.
2. The Josephson energy couples only neighbouring number states.
3. The Hamiltonian therefore has a tridiagonal matrix structure, making numerical diagonalisation particularly efficient.

In the next section we shall exploit this simple matrix structure to compute the energy levels and eigenstates of the transmon numerically.

3.3.4 Numerical Diagonalisation of the Transmon

Having written the transmon Hamiltonian in both the phase and number representations, we are now ready to determine its energy spectrum. Unlike the harmonic oscillator, the cosine potential prevents us from finding a simple closed-form solution. Instead, we shall solve the problem numerically using exactly the same philosophy developed earlier in this chapter.

The basic idea is straightforward. We first represent the Hamiltonian as a finite matrix, and then numerically diagonalise that matrix to obtain its eigenvalues and eigenvectors.

Why use the number basis?

Although the phase representation, Eq. (255), could be solved using finite-difference methods, there is an even simpler approach.

In the number basis, Eq. (261), the Hamiltonian is already expressed as a matrix acting on the states

$$\{|N\rangle\} = \{\ldots, |-2\rangle, |-1\rangle, |0\rangle, |1\rangle, |2\rangle, \ldots\}. \tag{262}$$

The charging energy is diagonal,

$$\langle N|H_C|N'\rangle = 4E_C N^2 \delta_{NN'}, \tag{263}$$

while the Josephson term couples only neighbouring number states,

$$\langle N|H_J|N'\rangle = -\frac{E_J}{2}\left(\delta_{N,N'+1} + \delta_{N,N'-1}\right). \tag{264}$$

Consequently, the Hamiltonian matrix is tridiagonal,

$$H = \begin{pmatrix} \ddots & \ddots & & & \\ -\frac{E_J}{2} & 4E_C & -\frac{E_J}{2} & & \\ & -\frac{E_J}{2} & 0 & -\frac{E_J}{2} & \\ & & -\frac{E_J}{2} & 4E_C & -\frac{E_J}{2} \\ & & & \ddots & \ddots \end{pmatrix}. \tag{265}$$

Only neighbouring number states communicate through Josephson tunnelling, making this Hamiltonian particularly efficient to solve numerically.

Basis truncation

The Hamiltonian above is formally infinite-dimensional because there are infinitely many possible values of N.

Fortunately, the low-energy eigenstates occupy only a finite range of number states. We may therefore truncate the Hilbert space,

$$N = -N_{\max}, \ldots, N_{\max}, \tag{266}$$

giving a matrix of dimension

$$d = 2N_{\max} + 1. \tag{267}$$

For sufficiently large $N_{\max}$, the low-energy eigenvalues become independent of the truncation size. In practice one simply increases $N_{\max}$ until the desired energy levels converge.

This idea is common throughout computational quantum mechanics. Rather than attempting to solve an infinite-dimensional problem exactly, we replace it by a finite matrix whose dimension is increased until the results no longer change.

Numerical diagonalisation

Once the Hamiltonian matrix has been constructed, finding the energy levels is conceptually no different from the examples studied earlier in this chapter.

We solve the eigenvalue equation

$$H|\psi_n\rangle = E_n|\psi_n\rangle, \tag{268}$$

where

$$|\psi_n\rangle = \sum_N c_N^{(n)}|N\rangle. \tag{269}$$

The coefficients $c_N^{(n)}$ are obtained directly from the numerical eigenvectors returned by the diagonalisation routine.

The eigenvalues

$$E_0, E_1, E_2, \ldots \tag{270}$$

give the allowed energies of the transmon, while the corresponding eigenvectors determine the probability of finding each number state in a given energy eigenstate.

3.3.5 The Harmonic Limit

Although the transmon Hamiltonian cannot generally be solved analytically, considerable insight may be obtained by considering the experimentally important limit

$$E_J \gg E_C. \tag{271}$$

Physically, this corresponds to strong Josephson tunnelling, causing the superconducting phase to fluctuate only weakly around the minimum of the cosine potential.

The cosine may therefore be expanded about

$$\phi = 0, \tag{272}$$

giving

$$\cos\phi = 1 - \frac{\phi^2}{2} + \frac{\phi^4}{24} + \cdots. \tag{273}$$

Substituting this into the Hamiltonian,

$$H = 4E_C N^2 - E_J\left(1 - \frac{\phi^2}{2} + \frac{\phi^4}{24} + \cdots\right)$$

$$= 4E_C N^2 + \frac{E_J}{2}\phi^2 - \frac{E_J}{24}\phi^4 - E_J + \cdots . \tag{274}$$

Ignoring the constant energy shift, the leading-order Hamiltonian is

$$H_{\mathrm{HO}} = 4E_C N^2 + \frac{E_J}{2}\phi^2. \tag{275}$$

This is simply the Hamiltonian of a quantum harmonic oscillator.

The correspondence with ordinary quantum mechanics is

Harmonic oscillator	Transmon
x	ϕ
p	N
$\frac{p^2}{2m}$	$4E_C N^2$
$\frac{1}{2}m\omega^2 x^2$	$\frac{E_J}{2}\phi^2$

Matching coefficients gives the plasma frequency

$$\omega_p = \frac{\sqrt{8E_C E_J}}{\hbar}, \tag{276}$$

which sets the characteristic energy scale of the transmon.

3.3.6 Anharmonicity

If the transmon were exactly harmonic, its energy levels would be equally spaced,

$$E_n = \hbar\omega_p \left(n + \frac{1}{2}\right). \tag{277}$$

This would make it impossible to isolate just two quantum states.

To see why, imagine applying microwave radiation resonant with the transition

$$|0\rangle \to |1\rangle. \tag{278}$$

Because every neighbouring pair of levels is separated by the same energy, exactly the same microwave field would also drive

$$|1\rangle \to |2\rangle, \tag{279}$$

allowing the system to leave the desired qubit subspace.

Fortunately, the cosine potential is not exactly quadratic. The quartic term,

$$-\frac{E_J}{24}\phi^4, \tag{280}$$

slightly shifts the excited-state energies, making the level spacing unequal.

This departure from equally spaced energy levels is known as the *anharmonicity* of the transmon.

Although the anharmonicity is relatively small, it is sufficient to allow microwave pulses to selectively address the lowest transition, forming an effective two-level quantum system.

The transmon therefore occupies an interesting middle ground. It is close enough to a harmonic oscillator that it inherits excellent coherence properties, yet sufficiently anharmonic that the lowest two levels may be isolated and used as a qubit.

In the next section we shall see that these two levels naturally define the computational basis of a superconducting qubit and provide the starting point for describing its interaction with electromagnetic fields.

Exercises

1. **Numerical diagonalisation of the transmon.**
   Construct the transmon Hamiltonian

$$H = 4E_C\hat{N}^2 - E_J\cos\phi \tag{281}$$

   in the number basis,

$$H = 4E_C\sum_N N^2|N\rangle\langle N| - \frac{E_J}{2}\sum_N\left(|N\rangle\langle N+1| + |N+1\rangle\langle N|\right). \tag{282}$$

   To do so:

   (a) Choose a basis truncation $N = -N_{\max}, \ldots, N_{\max}$, where $N_{\max}$ is sufficiently large.
   (b) Construct the corresponding Hamiltonian matrix.
   (c) Numerically diagonalise the Hamiltonian.
   (d) Plot the lowest six eigenvalues.
   (e) Increase $N_{\max}$ until the lowest four eigenvalues no longer change appreciably. What value of $N_{\max}$ is sufficient for convergence?

2. **Wavefunctions in the number basis.**
   The numerical diagonalisation returns both the eigenvalues and eigenvectors of the Hamiltonian.

   (a) Plot the coefficients $|c_N^{(n)}|^2$ for the four lowest-energy eigenstates,

$$|\psi_n\rangle = \sum_N c_N^{(n)}|N\rangle.$$

   Interpret these coefficients as the probability of finding N transferred Cooper pairs.

(b) How do these probability distributions change as the ratio E_J/E_C is increased?

(c) Explain physically why increasing E_J/E_C causes the low-energy eigenstates to become spread over a larger range of number states.

(d) Compare your numerical results with the harmonic approximation discussed in the text. Under what conditions does the harmonic approximation accurately describe the lowest-energy states?

Transition frequencies and anharmonicity.

Using your numerical eigenvalues, compute the transition frequencies

$$\omega_{01} = \frac{E_1 - E_0}{\hbar}, \qquad \omega_{12} = \frac{E_2 - E_1}{\hbar}. \tag{283}$$

1. Plot both transition frequencies as a function of the ratio E_J/E_C.
2. Show that the two frequencies become equal in the harmonic limit.
3. Define the anharmonicity

   $$\alpha = \omega_{12} - \omega_{01},$$

   and plot α as a function of E_J/E_C.
4. Explain why a non-zero anharmonicity is essential for operating the transmon as a qubit.